

Exercise 1

(a) What is the value of $\bar{y}_u$?

The population values are

$$y_1 = 98, \quad y_2 = 102, \quad y_3 = 154, \quad y_4 = 133, \quad y_5 = 190, \quad y_6 = 175.$$

The population mean is

$$\bar{y}_U = \frac{1}{N} \sum_{i=1}^N y_i = \frac{98 + 102 + 154 + 133 + 190 + 175}{6} = \frac{852}{6} = 142.$$

Hence,

$$\boxed{\bar{y}_U = 142}.$$

(b) Comparison of Sampling Plans

Plan 2

Each sample has probability $1/8$.

Sample S	$P(S)$	$\bar{y}_S$
$\{1, 3, 5\}$	$1/8$	$\frac{98 + 154 + 190}{3} = 147.333$
$\{1, 3, 6\}$	$1/8$	$\frac{98 + 154 + 175}{3} = 142.333$
$\{1, 4, 5\}$	$1/8$	$\frac{98 + 133 + 190}{3} = 140.333$
$\{1, 4, 6\}$	$1/8$	$\frac{98 + 133 + 175}{3} = 135.333$
$\{2, 3, 5\}$	$1/8$	$\frac{102 + 154 + 190}{3} = 148.667$
$\{2, 3, 6\}$	$1/8$	$\frac{102 + 154 + 175}{3} = 143.667$
$\{2, 4, 5\}$	$1/8$	$\frac{102 + 133 + 190}{3} = 141.667$
$\{2, 4, 6\}$	$1/8$	$\frac{102 + 133 + 175}{3} = 136.667$

(i) Expected Value

$$E(\bar{y}_S) = \sum_{k=1}^8 P(S_k) \bar{y}_{S_k} = \frac{1}{8} \sum_{k=1}^8 \bar{y}_{S_k}.$$

Thus,

$$E(\bar{y}_S) = \frac{1}{8}(147.333+142.333+140.333+135.333+148.667+143.667+141.667+136.667).$$

Therefore,

$$E(\bar{y}_S) = 142.$$

Hence,

$$\boxed{E(\bar{y}_S) = 142}.$$

(ii) Variance

Using the definition

$$\text{Var}(\bar{y}_S) = E \left[(\bar{y}_S - E(\bar{y}_S))^2 \right],$$

and since $E(\bar{y}_S) = 142$,

$$\text{Var}(\bar{y}_S) = \frac{1}{8} \sum_{k=1}^8 (\bar{y}_{S_k} - 142)^2.$$

Hence,

$$\begin{aligned} \text{Var}(\bar{y}_S) = \frac{1}{8} & \left[(147.333 - 142)^2 + (142.333 - 142)^2 \right. \\ & + (140.333 - 142)^2 + (135.333 - 142)^2 \\ & + (148.667 - 142)^2 + (143.667 - 142)^2 \\ & \left. + (141.667 - 142)^2 + (136.667 - 142)^2 \right]. \end{aligned}$$

Therefore,

$$\text{Var}(\bar{y}_S) = 18.94.$$

Hence,

$$\boxed{\text{Var}(\bar{y}_S) = 18.94}.$$

(iii) Bias

$$\text{Bias}(\bar{y}_S) = E(\bar{y}_S) - \bar{y}_U = 142 - 142 = 0.$$

Hence,

$$\boxed{\text{Bias}(\bar{y}_S) = 0}.$$

(iv) Mean Squared Error

$$\text{MSE}(\bar{y}_S) = \text{Var}(\bar{y}_S) + \text{Bias}(\bar{y}_S)^2.$$

Thus,

$$\text{MSE}(\bar{y}_S) = 18.94 + 0^2 = 18.94.$$

Hence,

$$\boxed{\text{MSE}(\bar{y}_S) = 18.94}.$$

Plan 1

Sample S	$P(S)$	$\bar{y}_S$
$\{1, 4, 6\}$	$1/4$	$\frac{98 + 133 + 175}{3} = 135.333$
$\{2, 3, 6\}$	$1/2$	$\frac{102 + 154 + 175}{3} = 143.667$
$\{1, 3, 5\}$	$1/4$	$\frac{98 + 154 + 190}{3} = 147.333$

(i) Expected Value

$$E(\bar{y}_S) = \sum_{k=1}^3 P(S_k) \bar{y}_{S_k}.$$

Hence,

$$E(\bar{y}_S) = \frac{1}{4}(135.333) + \frac{1}{2}(143.667) + \frac{1}{4}(147.333).$$

Therefore,

$$E(\bar{y}_S) = 142.5.$$

Hence,

$$\boxed{E(\bar{y}_S) = 142.5}.$$

(ii) Variance

Using

$$\text{Var}(\bar{y}_S) = E \left[(\bar{y}_S - E(\bar{y}_S))^2 \right],$$

and since $E(\bar{y}_S) = 142.5$,

$$\text{Var}(\bar{y}_S) = \sum_{k=1}^3 P(S_k) (\bar{y}_{S_k} - 142.5)^2.$$

Thus,

$$\begin{aligned}\text{Var}(\bar{y}_S) &= \frac{1}{4}(135.333 - 142.5)^2 \\ &\quad + \frac{1}{2}(143.667 - 142.5)^2 \\ &\quad + \frac{1}{4}(147.333 - 142.5)^2.\end{aligned}$$

Therefore,

$$\text{Var}(\bar{y}_S) = 19.361.$$

Hence,

$$\boxed{\text{Var}(\bar{y}_S) = 19.361}.$$

(iii) Bias

$$\text{Bias}(\bar{y}_S) = E(\bar{y}_S) - \bar{y}_U = 142.5 - 142 = 0.5.$$

Hence,

$$\boxed{\text{Bias}(\bar{y}_S) = 0.5}.$$

(iv) Mean Squared Error

$$\text{MSE}(\bar{y}_S) = \text{Var}(\bar{y}_S) + \text{Bias}(\bar{y}_S)^2.$$

Thus,

$$\text{MSE}(\bar{y}_S) = 19.361 + (0.5)^2 = 19.361 + 0.25 = 19.611$$

Hence,

$$\boxed{\text{MSE}(\bar{y}_S) = 19.611}.$$

Comparison of Sampling Plans

Quantity	Plan 1	Plan 2
$E(\bar{y}_S)$	142	142.5
$\text{Var}(\bar{y}_S)$	18.94	19.361
$\text{Bias}(\bar{y}_S)$	0	0.5
$\text{MSE}(\bar{y}_S)$	18.94	19.61

Plan 1 is unbiased and a smaller variance and thus a smaller MSE. In statistics, a plan that minimizes MSE is often preferred because it represents a better trade-off between accuracy and precision.

Exercise 2: Household Wastewater Survey

(a) SRS vs SRSWR

Answer: This is equivalent to **Simple Random Sampling Without Replacement (SRS)**.

Justification: Even though a generator is used, the researcher *ignores* duplicates. This ensures the final sample consists of 3 unique units, which defines SRS.

(b) Probability of specific unordered sample $\mathcal{S} = \{5, 18, 72\}$

In SRS, every possible sample of size n has probability $P(\mathcal{S}) = 1/\binom{N}{n}$.

$$P(\mathcal{S}) = \frac{1}{\binom{80}{3}} = \frac{1}{82,160}$$

(c) Inclusion probability π_j

For SRS, the probability of any unit being included is $\pi_j = n/N$.

$$\pi_j = \frac{3}{80} = 0.0375$$

Exercise 3: Professor's Exam Audit

(a) Probability of checking a specific exam

This is sampling **with replacement** (SRSWR).

1. $P(\text{not selected in 1 draw}) = 1 - 1/100 = 0.99$.
2. $P(\text{never selected in 5 draws}) = (0.99)^5 \approx 0.951$.
3. $P(\text{selected at least once}) = 1 - 0.951 = 0.049$.

Conclusion: The student is **incorrect**. The linear 5/100 calculation only works for SRS. In SRSWR, the probability is slightly lower (≈ 0.049).

(b) Probability of exactly 4 unique exams

Scaffold steps:

- **Total Sequences:** $100^5 = 10,000,000,000$.

- **Choosing IDs:** $\binom{100}{1}$ ways to pick the "double" ID, and $\binom{99}{3}$ ways to pick the three "singles". This results in 15684900 ways to choose 1 ID and 3 unique IDs from the remaining IDs.
- **Orderings:** For a set with one repeat (e.g., $\{A, A, B, C, D\}$), there are $\frac{5!}{2!} = 60$ sequences.
- **Final Probability:** $\frac{100 \times \binom{99}{3} \times 60}{100^5} \approx 0.0941$.

(c) Replacement Strategy

Answer: She should have sampled **without replacement** (SRS).

Justification: In SRS, the inclusion probability is exactly $\pi_i = n/N = 5/100 = 0.05$.